The Proteasome Challenge

A Scientific Laboratory in Embedded-Wavefunction
Electronic Structure of Isolated MLN2238

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Abstract

The 20S proteasome is a molecular target of boronic-acid inhibitors including bortezomib and ixazomib. This article develops the electronic-structure methods of a computational laboratory focused on the *isolated* active ixazomib species MLN2238. It proceeds from a restricted Hartree–Fock reference through localized fragment construction, Schmidt and correlated bath orbitals, full configuration interaction and coupled-cluster solvers, selected-configuration quantum sampling, a mixed 42-fragment embedded-wavefunction campaign, and global energy reconstruction. The example uses 19 FCI, 12 CCSD, and 11 TrimSQD fragment assignments. Published figures and source equations connect each method to the primary literature, while distinguishing the present molecular calculation from published studies of Trp-cage and protein–ligand complexes. An isolated inhibitor electronic energy is not a measurement of proteasome binding or inhibition.

Keywords: proteasome; ixazomib; MLN2238; Hartree–Fock; embedded wavefunction; FCI; CCSD; sample-based quantum diagonalization; reduced density matrices.

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1 Biological and Chemical Context

1.1 A molecular machine for protein degradation

Cells continually synthesize, regulate, and degrade proteins. A central pathway for regulated protein turnover is the **ubiquitin–proteasome system**, in which proteins marked for degradation are processed by a multiprotein protease. Its catalytic core, the **20S proteasome**, is a barrel-like assembly of 28 subunits, arranged as four seven-membered rings: $\alpha_{1–7}$–$\beta_{1–7}$–$\beta_{1–7}$–$\alpha_{1–7}$. Regulatory particles can associate with this core to form the 26S proteasome. [BIO-1]

The catalytic chamber contains three principal active β-subunit types, distinguished by their substrate preferences: **β1** (caspase-like), **β2** (trypsin-like), and **β5** (chymotrypsin-like). These active sites are N-terminal threonine proteases. The β5 site is an important target for proteasome-inhibiting anticancer agents. [BIO-1, BIO-4]

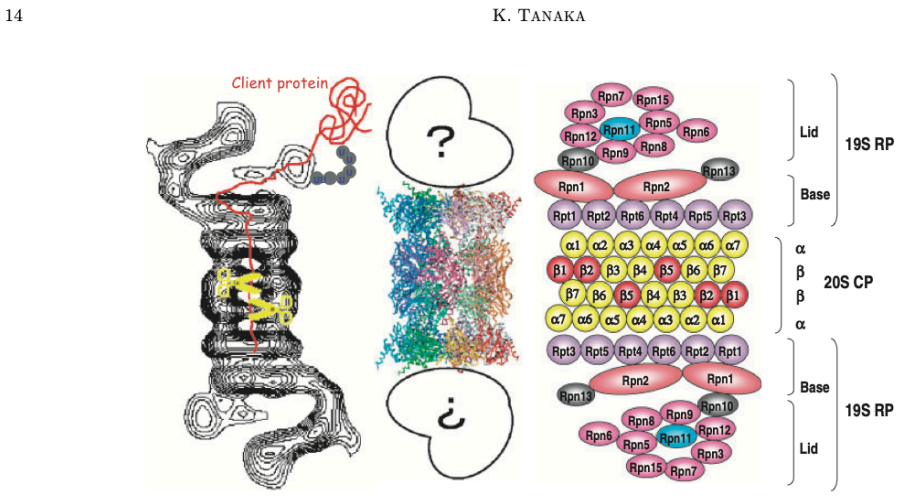


Fig. 1. Schematic diagram of the 26S proteasome. Left panel: Averaged image of the rat 26S proteasome complex based on electron micrographs. Photograph kindly provided by W. Baumeister. U, ubiquitin. Middle panel: The overall tertiary structure of the bovine 20S proteasome (central portion); the structures of the 19S RPs have not yet been determined (the pair of symmetrically disposed terminal structures depicted by question marks). Right panel: Schematic drawing of the subunit structure. CP, core particle (20S proteasome); RP, 19S regulatory particle consisting of the base and lid subcomplexes; Rpn, RP non-ATPase; Rpt, RP triple-ATPase.

Figure 1: Schematic of the 26S proteasome, showing its electron-microscopy-derived architecture, the 20S core particle, and the regulatory and catalytic subunit organization. Tanaka (2009), original Fig. 1, p. 14. DOI: 10.2183/pjab.85.12. [BIO-1]

View the original published Figure 1 in the article PDF (printed p. 14, PDF p. 3).

1.2 Two inhibitors, a common catalytic target

Early medicinal-chemistry work established **dipeptidyl boronic acids** as potent proteasome inhibitors, with markedly greater potency than the peptide aldehydes examined in the same series. This chemistry contributed to the development of **bortezomib**. [BIO-2]

A later inhibitor, **ixazomib**, provides a closely related but chemically distinct example. In the nomenclature of the early pharmacological literature, **MLN9708** converts in aqueous media or plasma to **MLN2238**, the biologically active boronic-acid form. The electronic-structure system considered here is **isolated MLN2238**, rather than its precursor or a drug formulation. [BIO-3]

Bortezomib and MLN2238 both contain a boronic-acid functional group and preferentially inhibit the β5 catalytic site. The differences between their molecular structures can be examined using the chemical identities reported for these inhibitors. The present calculation examines the electronic structure of isolated MLN2238. **Lee et al. (2011), Fig. 1** identifies MLN9708 and its active form MLN2238; the panel does not depict bortezomib. [BIO-3, BIO-6, CHEM-1]

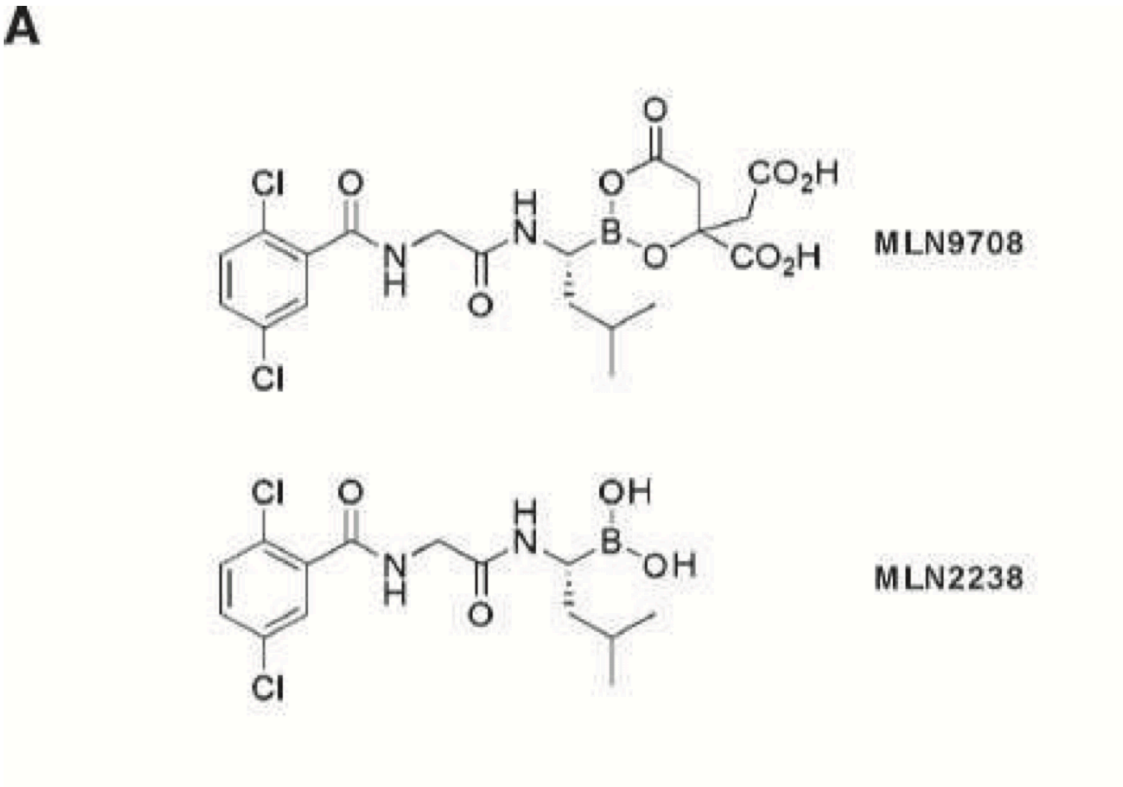


Figure 2: Chemical structures of the MLN9708 boronate ester precursor (top) and active boronic-acid MLN2238 (bottom). Lee et al. (2011), original Fig. 1, *Clinical Cancer Research* **17**, 7313–7323. DOI: 10.1158/1078-0432.CCR-11-0636. [BIO-6]

In biochemical assays reported by Kupperman and colleagues, the measured **β5 inhibition constants** were **0.55 nM for bortezomib** and **0.93 nM for MLN2238**. Their **β5 dissociation half-lives** were more widely separated: **110 minutes** and **18 minutes**, respectively. The reported ranges of the inhibition constants overlap; these measurements do not establish a precisely resolved free-energy difference between the two compounds. [BIO-3, Table 1]

The biochemical inhibition constants and dissociation half-lives in Kupperman et al., Table 1, describe enzyme–inhibitor interactions. An isolated MLN2238 electronic energy is a

distinct molecular quantity. [BIO-3, Table 1]

1.3 Structural evidence from human proteasomes

High-resolution X-ray crystallography provides an atomic starting point for studying the inhibitors in their biological environment. Schrader and colleagues reported human 20S proteasome structures bound to several inhibitors, including **bortezomib (PDB 5LF3, 2.10 Å)** and **ixazomib (PDB 5LF7, 2.00 Å)**. The deposited inhibitor complexes distinguish the broader structural research question from the smaller electronic-structure experiment carried out here. [BIO-4, STRUCT-1, STRUCT-2]

These structures motivate future calculations on the β5 active site and on inhibitor–protein interactions. **They are not themselves the computational systems used in the present 42-fragment exercise.** In particular, a structure of an inhibitor in a proteasome pocket does not mean that an isolated-molecule energy calculation can be interpreted as a binding-energy calculation. [BIO-4; Challenge scope]

1.4 Electronic structure of isolated ixazomib

The electronic-structure calculation concerns the **active, isolated ixazomib molecule MLN2238**. Its molecular formula is $\mathbf{C_{14}H_{19}BCl_2N_2O_4}$. For the neutral molecule, this corresponds to **188 electrons**. The experiment begins with a restricted Hartree–Fock description in the STO-3G basis and then examines how embedded wavefunction methods and quantum-sampled electronic configurations recover correlation beyond that reference. [CHEM-1; Challenge specification]

The present demonstration will calculate an electronic-structure reference and embedded correlated fragment quantities for this isolated molecule. Its outputs are **not** the binding or inhibition measurements reported in the papers above. [Challenge model definition]

The electronic-structure calculation begins with a Hartree–Fock description of the electrons at the selected nuclear geometry. This mean-field reference provides the starting point for the correlated calculations.

2 The Hartree–Fock Reference

2.1 The many-electron description

The molecular electronic-structure problem begins with the Schrödinger equation for the electrons in the field of a fixed nuclear geometry. The Hamiltonian contains electron kinetic energy, electron–nuclear attraction, and electron–electron repulsion. The last term couples the motions of the electrons, making an exact many-electron description difficult. [QC-1, QC-3, §III]

Bartlett and Musiał write the Born–Oppenheimer Hamiltonian as their **Eq. (1)**, with the alternative partitioned form given in their **Eq. (2)**:

$$H(\mathbf{R}) = \sum_i h(\mathbf{r}_i) + \frac{1}{2}\sum_{i,j} \hat{g}(\mathbf{r}_i, \mathbf{r}_j) + E_{NN}.$$

This is **Eq. (2) from the source** in the authors' form: one-electron terms, electron–electron interactions, and a nuclear–nuclear energy E_{NN}, constant for fixed geometry. [QC-3, §III, Eqs. (1)–(2), journal pp. 294–295]

2.2 The Hartree–Fock approximation

Roothaan's molecular-orbital treatment approximates the many-electron wavefunction using an antisymmetrized product of occupied spin orbitals—a single Slater determinant. The orbitals are optimized variationally within that restricted form, producing a self-consistent mean-field reference. [QC-1, §§I–III]

A finite atomic-orbital basis expands each molecular orbital. The resulting self-consistent equations become a generalized matrix eigenvalue problem:

$$\mathbf{FC} = \mathbf{SC}\varepsilon.$$

Roothaan defines **F** as the matrix appearing in the self-consistent molecular-orbital equations, **S** as the basis overlap matrix, and **C** as the coefficient matrix. His Eq. (58′) calls **ε a Hermitian matrix**, not yet necessarily a diagonal list of orbital energies. The self-consistent-field equations are then solved iteratively. [QC-1, printed p. 76, text immediately preceding Eq. (58′); p. 77, SCF procedure]

In the same way as in the previous section it follows that the ϵ_{ij}'s are the elements of an hermitian matrix $\boldsymbol{\varepsilon}$; the two equations (58) then become equivalent. We may write then for Eqs. (58)

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}. \qquad (58')$$

Figure 3: Roothaan (1951), Eq. (58′), printed p. 76. Generalized molecular-orbital matrix equation, with ε described as Hermitian. DOI: 10.1103/RevModPhys.23.69.

2.3 Why restricted Hartree–Fock?

For a closed-shell molecule, restricted Hartree–Fock (RHF) occupies each filled spatial orbital with two electrons of opposite spin. This differs from unrestricted and restricted open-shell reference formulations, which allow alternative spin-orbital occupation descriptions. [QC-3,

§V.B, Fig. 7]

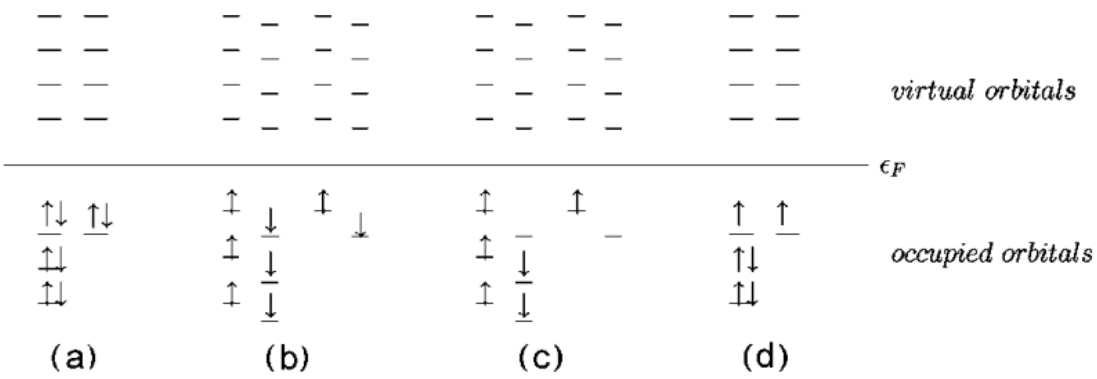


FIG. 7. The choices made for the spin-orbital form: (a) RHF (closed-shell molecule with doubly occupied spatial orbitals), (b) UHF (closed-shell spin polarized situation), (c) UHF open-shell triplet, and (d) ROHF (triplet maximum double occupancy).

Figure 4: Bartlett and Musiał (2007), original Fig. 7 (printed p. 305). Panel (a) illustrates the doubly occupied RHF reference; panels (b)–(d) provide alternatives. DOI: 10.1103/RevModPhys.79.291.

2.4 The MLN2238 molecular reference

For the selected isolated MLN2238 model, the Challenge's restricted Hartree–Fock calculation uses the **STO-3G** minimal basis. The established reference contains **188 electrons** and **142 spatial atomic-orbital basis functions**. For a closed-shell RHF solution, this corresponds to **94 doubly occupied spatial orbitals** and **48 unoccupied spatial orbitals**. [Challenge reference specification]

When the Stage 1 calculation is displayed, inspect the validated total Hartree–Fock energy, orbital occupations, and convergence record. The energy is defined for this *particular nuclear geometry, charge, basis, and Hamiltonian*. It cannot be interpreted as a binding energy by itself. [QC-1; Challenge interpretation]

2.5 Why Hartree–Fock is not the final answer

The self-consistent solution optimizes one Slater determinant but does not allow the independent electronic configurations needed to capture the full correlated many-electron state. Bartlett and Musiał next present the configuration-interaction and coupled-cluster constructions in §§III.A and III.C. [QC-3, §III.A and §III.C]

The Hartree–Fock reference supplies the starting electronic description for fragment and

bath construction in the embedding treatment of Nusspickel and Booth (2022), §II. [QC-4, §II]

3 Embedded Wavefunction Fragmentation and Bath Construction

3.1 The locality question

The Hartree–Fock calculation of **isolated MLN2238** produces a mean-field description in a common molecular-orbital basis. Correlated electronic-structure methods extend this single-determinant reference. A correlated calculation over the full molecular orbital space would need to describe excitations involving many electrons and orbitals simultaneously; the size of the possible many-electron configuration space increases rapidly with the number of orbitals. Instead, we seek a description of correlation near a localized region of the molecule while still retaining its interaction with the surrounding electronic environment. [QC-4, §I and §II; QC-7, §2.1]

Embedded wavefunction (EWF) theory addresses this problem by constructing smaller *fragment-plus-bath* electronic spaces. The fragment selects the local orbitals of interest; the bath represents relevant degrees of freedom of the rest of the molecule. The bath is essential: an isolated fragment cut out of the molecule would generally lose the electronic coupling to its surroundings that affects its correlated state. [QC-4, §I–II; QC-7, §2.1]

The scientific question in this stage is therefore not simply, *How many pieces can we divide the molecule into?* It is **which environmental orbitals must accompany each piece to retain the electronic effects that matter.**

3.2 Localized fragments: intrinsic atomic orbitals

Canonical Hartree–Fock molecular orbitals describe the whole molecule and need not belong to individual chemical regions. To identify local correlated problems, the EWF construction uses **intrinsic atomic orbitals (IAOs)**: atom-associated localized orbitals that provide a chemically interpretable representation of the occupied mean-field space. These IAOs define the fragment subspaces without treating each fragment as a free-standing molecule. [QC-4, §II.A, journal p. 011046-2; QC-7, §2.1.1, preprint p. 8]

Nusspickel and Booth define atomic fragments in **§II.A** and construct the environment bath from the Hartree–Fock reference using the DMET/Schmidt procedure. The local atom-associated fragment definition is taken from their construction rather than from a new orbital-space notation introduced here. [QC-4, §II.A, journal p. 011046-2]

3.3 The Schmidt bath: retaining mean-field entanglement

The **Schmidt decomposition** of the Hartree–Fock reference across a fragment–environment partition identifies a compact set of environment orbitals coupled to the fragment. In density-

matrix embedding theory (DMET), these become the **entanglement bath orbitals**. The initial DMET cluster comprises the fragment orbitals together with the Schmidt bath described in **Nusspickel and Booth, §II.A** and **Shajan et al., §2.1.1**. No additional cluster-space equation is introduced here. [QC-4, §II.A; QC-7, §2.1.1]

The key mathematical property is that the DMET bath can reproduce the **mean-field one-particle density matrix on the fragment** when the reference state is restricted to this cluster. The number of entanglement bath orbitals is bounded by the number of fragment orbitals, so the mean-field bath remains compact even as the total system becomes large. That is a statement about the **Hartree–Fock reference**; it does **not** mean the same bath is sufficient to recover all correlation effects. [QC-4, §II.A–B, journal p. 011046-2; QC-7, §2.1.1]

3.4 Why the mean-field bath must be expanded

Electron correlation introduces fluctuations and excitations that the Hartree–Fock determinant does not contain. An environment orbital that contributes little to the *mean-field* fragment density can nonetheless be important when electrons excite from occupied to virtual states. Consequently, EWF augments the DMET bath with additional **interacting bath natural orbitals**, often abbreviated **BNOs**. [QC-4, §II.B, journal pp. 011046-2–3; QC-7, §2.1.2, preprint pp. 8–9]

Nusspickel and Booth construct the relevant information using **two restricted second-order Møller–Plesset (MP2) calculations** for each cluster. In one, the occupied excitation indices are restricted to the initial fragment-plus-DMET-bath space; in the other, the virtual indices are restricted. The resulting approximate correlated one-particle density matrices, projected into the environment, are diagonalized to obtain an ordered set of additional occupied and virtual bath orbitals. This turns MP2 into an **orbital-selection guide**: MP2 is used here to identify where the correlated space should be enlarged, not to declare the final fragment energy. [QC-4, §II.B, Eqs. (1a–1b), (2a–2b); QC-7, §2.1.2]

The **bath natural-orbital threshold** η selects the extra environment orbitals in Nusspickel and Booth, **§II.B**. In the authors' words, unoccupied-orbital occupation numbers must be **larger than** η, while occupied-orbital occupation numbers must be **smaller than** $2 - \eta$. These occupation-number conditions control inclusion of the interacting bath orbitals. [QC-4, §II.B, journal p. 011046-3; QC-7, §2.1.2]

A smaller η retains more additional orbitals, producing a larger correlated problem. A larger η truncates the bath more aggressively. This gives a direct scientific tradeoff: cluster size and correlated-solver cost versus the degree of environmental correlation represented. The existence of a systematic threshold is not a claim that a particular finite threshold is already converged; convergence must be investigated for the property being studied. [QC-4, §II.B and Fig. 1–3]

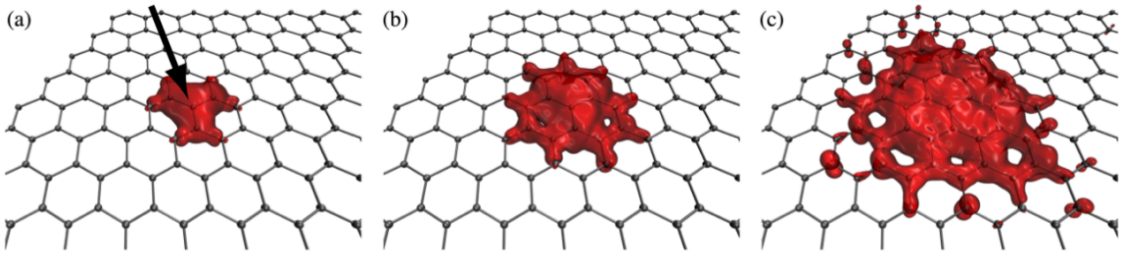


FIG. 1. Isosurface of the cluster orbital density in the embedding of a carbon atom [marked by the black arrow in (a)] in graphene. (a)–(c) The cluster grows in size as the bath orbital threshold η is reduced from 10^{-4} (a) to 10^{-6} (b) and 10^{-8} (c). An isovalue of 0.2 is used for all plots.

Figure 5: Original **Fig. 1** from Nusspickel and Booth (2022), **Physical Review X 12**, 011046, printed p. 011046-3. The red cluster-orbital-density isosurfaces around a carbon atom in **graphene** grow as η decreases from 10^{-4} (a), to 10^{-6} (b), to 10^{-8} (c). These are **published graphene examples illustrating the embedding principle**, not images of MLN2238 or of a proteasome active site. DOI: 10.1103/PhysRevX.12.011046.

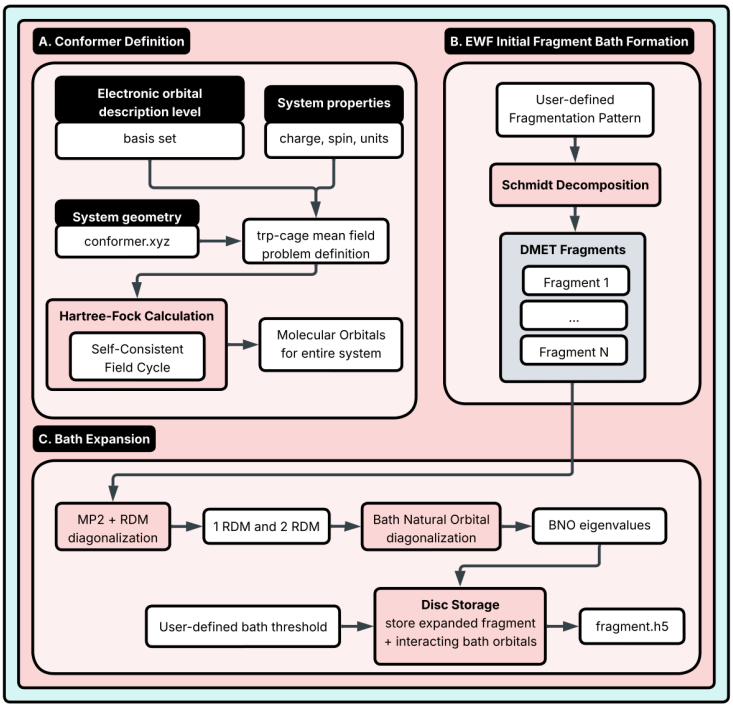


Figure 2: Schematic overview of the classical pre-processing and EWF workflow. The diagram illustrates (A) conformer definition and mean-field problem setup, (B) initial EWF fragment and DMET bath construction from the Hartree–Fock reference, and (C) MP2-level interacting bath expansion via bath natural orbitals.

Figure 6: Original **Figure 2** from Shajan et al. (2026), "Molecular Quantum Computations on a Protein," arXiv:2512.17130v2, PDF/printed p. **12**. The published schematic connects (A) Hartree–Fock preparation for a **Trp-cage** conformer to (B) Schmidt/DMET bath formation and (C) MP2-informed bath-natural-orbital expansion. This schematic comes from the **published Trp-cage preparation**, not the isolated MLN2238 calculation.

The source diagram makes the separation between the two baths explicit: the Schmidt decomposition follows the mean-field reference, whereas the MP2-informed step expands the correlated cluster beyond its initial DMET bath. The `fragment.h5` label identifies the result of the **published** preparation workflow; it is not an instruction to change the file handling or execution of this Challenge. [QC-7, §§2.1.1–2.1.2 and original Fig. 2]

3.5 The scientific output: a distribution of correlated cluster sizes

Every fragment is associated with an EWF cluster, but different local chemical environments need not produce equally large clusters. The number of retained orbitals depends on the fragment, its coupling to the mean-field environment, the correlation-informed BNO occupations, and the selected bath threshold. As a result, **fragment count and correlated problem size are different quantities**. [QC-4, §II.B–C; QC-7, §2.1.2 and §3.2.1]

In this Challenge, the established construction yields **42 EWF cluster problems** for the

isolated MLN2238 model. Treat this as a **Challenge-specific measured or validated result**, not a number predicted in the cited papers. The scientific outputs to inspect are the identities of the generated fragments, each cluster's orbital count and occupation/electron information, and the resulting distribution of embedded problem sizes. These outputs will explain why different solvers are used later. [Challenge implementation/results; QC-7, §3.2.1 for the analogous solver-size rationale]

The distribution of cluster sizes identifies fragments whose electronic environments require larger or smaller interacting baths. Cluster size reflects the dimension of the embedded orbital space; it does not by itself establish a fragment's influence on an inhibitor–proteasome binding observable. [QC-4, §II.B–C]

3.6 What embedding has—and has not—solved

Embedding does not eliminate the electronic correlation problem. It reorganizes that problem into smaller, **interacting** fragment-plus-bath spaces suitable for correlated electronic-structure solvers. The final energy is not obtained by blindly adding complete energies from overlapping clusters; an appropriate partition or reconstruction of fragment contributions is needed to avoid double counting. Nusspickel and Booth express their original approach in terms of a Hartree–Fock reference plus partitioned correlation contributions,

$$E_{\mathrm{tot}} = E_{\mathrm{HF}} + E_{\mathrm{corr}} \approx E_{\mathrm{HF}} + \sum_{X} E^{X}_{\mathrm{corr}},$$

This is the published **Eq. (4)** (including the authors' definition of total energy as Hartree–Fock plus correlation), with the partitioning prescription given in §II.C. Global reconstruction methods combine correlated cluster information, including reduced density matrices and cumulants, to obtain whole-molecule expectation values. An individual fragment energy is not the total molecular energy. [QC-4, §II.C, Eq. (4); QC-7, §2.1; QC-5, cited for the later reconstruction stage]

The present system is **an isolated inhibitor**, with neither the proteasome binding site nor the second inhibitor included in the computed Hamiltonian. Thus the 42 embedded problems establish a route to studying **intramolecular electronic correlation** in MLN2238; they do not, on their own, predict proteasome inhibition constants or a comparative binding free energy. [Challenge model scope]

3.7 Correlated solutions of embedded clusters

The embedding construction produces smaller molecular Hamiltonians containing localized fragments and selected environmental orbitals. Their correlated many-electron solutions are calculated with **full configuration interaction (FCI)**, which expands over all determinants in a fixed cluster basis, or **coupled-cluster singles and doubles (CCSD)**, which represents correlation through connected excitation operators. The resulting classical baseline precedes

quantum-selected determinant subspaces. [QC-7, §3.2.1; QC-2, QC-3 for the subsequent methods]

4 Classical Correlation: Full Configuration Interaction and Coupled Cluster

4.1 Why a mean-field reference is not the final wavefunction

In Stage 1, the Hartree–Fock (HF) calculation represented **isolated MLN2238** by a single antisymmetrized product of molecular spin-orbitals. Stage 2 then transformed that reference into **42 embedded fragment-plus-bath electronic problems**. The embedded Hamiltonians still describe interacting electrons; embedding makes the problems smaller but does not solve their electron correlation. In this stage we ask: **how can each cluster's electronic state go beyond one determinant, and what is the computational price of doing so?** [QC-3, §III.A–C; QC-7, §3.2.1; Challenge-defined 42-cluster result]

An HF determinant accounts for fermionic antisymmetry and the average interaction experienced by each electron. It does not generally capture the correlated fluctuations in which the probability of finding one electron depends on where the others are found. Correlated wavefunction methods therefore give the electronic state more flexibility than a single determinant. Two established approaches to doing this are **full configuration interaction (FCI)** and **coupled-cluster singles and doubles (CCSD)**. [QC-3, §§I, III.A–C]

4.2 Excitations are the building blocks

Let $|\Phi_0\rangle$ denote the HF reference for a particular embedded cluster. A **singly excited determinant** replaces one occupied spin-orbital i by an unoccupied orbital a; a **double excitation** replaces two occupied orbitals i, j by two virtual orbitals a, b. Higher excitation ranks replace correspondingly more occupied orbitals. These are alternative many-electron **configurations in the same orbital basis**, not different molecular geometries or independent chemical fragments. [QC-3, §III.A, Fig. 2; QC-2, Introduction]

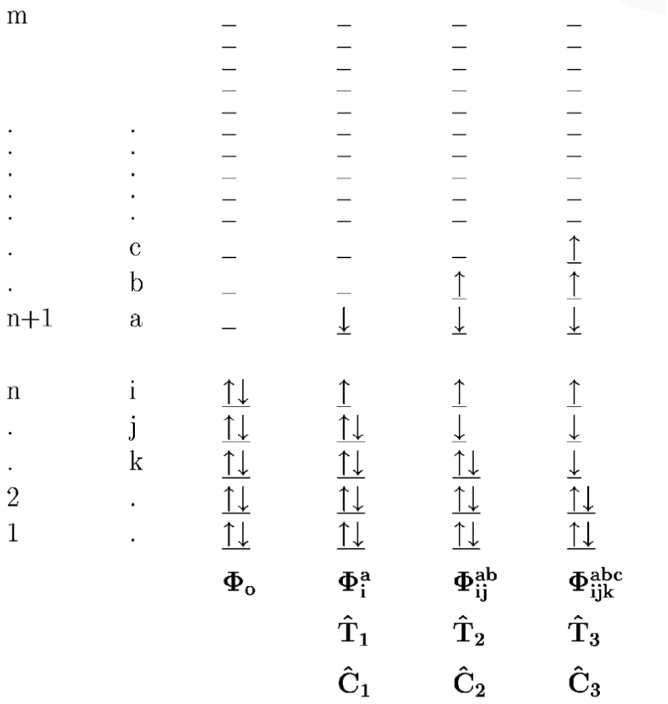


FIG. 2. Graphical examples of the selected single (Φ_i^a), double (Φ_{ij}^{ab}), and triple (Φ_{ijk}^{abc}) excitations due to the T_1 (C_1), T_2 (C_2), and T_3 (C_3) operators, respectively. Electrons from all possible occupied orbitals can be excited to all possible unoccupied orbitals, so $T_3\Phi_0=\sum_{\substack{i<j<k\\a<b<c}} t_{ijk}^{abc}\Phi_{ijk}^{abc}$.

Figure 7: Original **Fig. 2**, Bartlett and Musiał (2007), Reviews of Modern Physics **79**, 291–352, journal p. **293** (PDF p. 3). The diagram displays a reference determinant Φ_0 and illustrative single, double, and triple excitations, identified with the T_1/C_1, T_2/C_2, and T_3/C_3 operators. The figure describes **generic orbital excitations**, not a numerical result for MLN2238. DOI: 10.1103/RevModPhys.79.291.

This shared language of determinants will connect all three solver families encountered later: FCI includes the complete configuration space, CCSD builds a particular exponential wavefunction, and SQD will later retain a *selected* set of determinants inferred from quantum measurements. [QC-3, §§III.A–C; QC-6, Results]

4.3 Full configuration interaction: complete within a chosen space

Bartlett and Musiał describe full configuration interaction by including excitations through the full rank in a fixed finite basis. Their **Eq. (7)** writes the associated CI expansion using excitation operators:

$$\Psi_{\mathrm{CI}} = \left(1 + \sum_{p=1}^{n} \hat{C}_p\right)\Phi_0.$$

This is the authors' **Eq. (7)** and is explicitly linked to the determinant expansion in their **Eq. (6)**. For the MLN2238 calculation, the corresponding CI expansion applies to each **embedded cluster**, not the entire 188-electron molecule. The amplitudes c_I are found by solving the Hamiltonian eigenvalue problem within that finite space. FCI is exact **for the specified cluster Hamiltonian and finite basis**, provided the desired eigenpair is converged. It is **not** an exact solution of the complete isolated molecule, the enzyme–inhibitor complex, or the basis-set limit. [QC-3, §III.A, Eq. (6), journal p. 295; QC-2, Introduction]

Bartlett and Musiał note directly that the number of FCI determinants grows rapidly with the number of basis functions (their **§III.A, immediately after Eqs. (6)–(7)**). This growth limits the sizes of the spaces for which full determinant-based diagonalization is practical. [QC-3, §III.A, printed p. 295; QC-2, Introduction]

Knowles and Handy's determinant-based formulation makes the computational task explicit: rather than constructing and densely diagonalizing a full matrix, an iterative eigensolver repeatedly evaluates a Hamiltonian–vector product. Their **Eq. (1)** writes the Hamiltonian action as

$$a_I = H_{IJ} c_J .$$

Here H_{IJ} is the Hamiltonian matrix element between expansion states, as defined by Knowles and Handy. Their 1984 paper describes the determinant and spin-string representations that make this operation efficient. Such algorithms greatly improve what is computationally feasible, but they do **not remove the combinatorial growth of FCI**. [QC-2, Introduction, Eq. (1), pp. 315–316; §2, pp. 316–319]

The converged FCI ground-state energy satisfies the Rayleigh–Ritz variational principle for the fixed Hamiltonian and orbital basis: an FCI eigenvalue is an upper bound to that Hamiltonian's exact ground-state energy, reaching equality when the full allowed space is solved. The relevant qualification is always **which Hamiltonian and orbital space** are being considered. [QC-3, §III.A, journal p. 295]

4.4 CCSD: an exponential representation of correlation

CCSD retains a single reference determinant but uses a different representation of its correlated state:

$$|\Psi_{\mathrm{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2} |\Phi_0\rangle .$$

The operator $\hat{T}_1$ promotes one electron from an occupied to a virtual spin-orbital, while $\hat{T}_2$ describes two-electron excitations. Their coefficients are the **cluster amplitudes**. Bartlett and Musiał derive the general exponential ansatz $e^{\hat{T}}|\Phi_0\rangle$ and show how the CCSD approximation retains **connected** singles and doubles while setting the higher connected operators, such as T_3 and T_4, to zero. [QC-3, §II, journal p. 294; §III.C, Eqs. (24)–(31), journal p. 297]

Bartlett and Musiał compare CI and CC operators explicitly in their **Eqs. (39)–(42)**. For

example, their **Eqs. (40)–(41)** read:

$$C_2 = T_2 + \frac{1}{2}T_1^2, \qquad C_3 = T_3 + T_1T_2 + \frac{1}{3!}T_1^3.$$

The authors use these expressions to show how products of lower-rank connected cluster operators enter higher-rank components of the wavefunction. In CCSD the independent connected T_3 and higher operators are omitted, whereas products involving T_1 and T_2 remain. [QC-3, §III.C, Eqs. (39)–(42), printed p. 298]

The authors present the projected CC equations in **Eqs. (35)–(37)**. Their energy formula, **Eq. (37)**, is

$$E = \langle\Phi_0|e^{-T}He^{T}|\Phi_0\rangle = \langle\Phi_0|\bar{H}|\Phi_0\rangle.$$

For the associated amplitude projections, refer to the original **Eq. (36)** rather than substituting a newly generalized indexed formula. [QC-3, §III.C, Eqs. (35)–(37), printed p. 297]

4.5 Why CCSD is useful—and why FCI remains the reference

The exponential, linked-cluster construction makes properly formulated CCSD **size-extensive**: for noninteracting subsystems, its energy scales additively. Truncated linear CI models such as CISD generally do not share that property. The authors describe CCSD work as scaling like n^2N^4, where n and N are their occupied and virtual orbital counts, respectively (printed p. 298). These are the reasons CCSD can serve as a practical correlated method for clusters too large for an FCI calculation. [QC-3, §§I–II, journal pp. 292–294; §III.C, journal pp. 297–298]

CCSD remains an approximation. Bartlett and Musiał report the differences between CCSD and finite-basis FCI correlation energies for several molecules and bond lengths in **Table I** (printed p. 299), and discuss single-reference method limitations later in the review (§VI). These comparisons depend on the electronic problems and reference conditions considered. [QC-3, Table I, printed p. 299; §VI]

Within a **matching embedded Hamiltonian**, FCI provides a finite-space reference for checking correlated approximations. For different embedded clusters, however, a difference between their raw total energies is not itself a controlled test of solver accuracy; electron numbers, nuclear/effective one-body terms, and orbital spaces may differ. Any solver comparison must use the **same cluster and the same Hamiltonian**, with a documented comparison quantity. [QC-3, §III.A; QC-7, §3.2.1; Challenge interpretation]

4.6 The Challenge's classical baseline

The Stage 2 construction supplies **42 embedded electronic problems** for isolated MLN2238. The established **Experiment A / Stage B classical baseline** routes **19 clusters to FCI** and **23 clusters to CCSD**. These numbers are **Challenge-specific workflow assignments**,

not results or thresholds reported by Knowles and Handy, Bartlett and Musiał, or Shajan et al. The allocation preserves tractable exhaustive solutions while obtaining a classical correlated treatment across the full fragment set. [Challenge-defined Stage B allocation; QC-7, §3.2.1 for the methodological precedent]

The classical baseline is characterized by each cluster's **orbital and electron counts**, its assigned **FCI or CCSD solver**, and its **computed correlated quantities**, including convergence indicators. The comparison of correlated energies is meaningful only for matching Hamiltonians and consistently converged solutions. [QC-3, §III.A–C; QC-7, §3.2.1]

The central observational question is **where an exhaustive correlated solution remains feasible, and where a less exhaustive representation becomes necessary**. Note that a fragment requiring more orbitals is not automatically the fragment of greatest biological importance: size is an electronic **cost proxy**, not a measured contribution to proteasome binding or inhibition. [QC-2, Introduction; QC-7, §3.2.1; Challenge interpretation]

4.7 Toward quantum-selected configuration spaces

In the later experiment, the Challenge retains **19 FCI clusters**, assigns **12 to CCSD**, and selects **11 candidates** for TrimSQD. The 11 candidates are the designated larger clusters, selected using the established $n_{\mathrm{orb}} \geqslant 20$ **orbital-count criterion**. This is a **size-based route**, not a demonstrated ranking by correlation difficulty or biochemical significance. The purpose of the present classical baseline is to provide a classical point of comparison and to motivate why the full determinant space is an increasingly expensive representation. [Challenge's established solver plan; QC-7, §3.2.1; QC-6, SQD method as the next scientific idea]

SQD will preserve the FCI-like **linear expansion in selected determinants** while using quantum-generated bitstrings and classical configuration recovery to decide which determinants enter the subspace. The selected-determinant and configuration-recovery methodology is developed by Robledo-Moreno et al. (2025). Likewise, the molecular energy obtained by combining fragment information requires a specific EWF **density-matrix/cumulant reconstruction**, addressed in Stage 4. **Adding raw cluster energies is not a valid substitute.** [QC-6, Results; QC-5, §§II.A–B; QC-7, §3.3]

The classical baseline provides correlated calculations for embedded fragments of **isolated MLN2238**. These are electronic-structure results, not measurements of proteasome–inhibitor binding energy, binding free energy, or experimental K_i.

5 Selecting Fragments for Quantum Treatment

5.1 Why different fragments require different solvers

The 42 embedded electronic problems constructed from **isolated MLN2238** are not all equally expensive to correlate. In the classical baseline, each fragment-plus-bath Hamiltonian was assigned either full configuration interaction (FCI) or coupled-cluster singles

and doubles (CCSD). The next choice concerns **which embedded problems enter the quantum-sampled correlated solver**. This is a *solver-selection* decision, not a new fragmentation procedure. [Challenge-defined Stage 2 and Experiment A; QC-7, §3.2.1]

Shajan et al. organize their EWF clusters by the **number of molecular orbitals** and select an SQD route for larger clusters (§3.2.1, Fig. 4). Bartlett and Musiał describe the increasing FCI configuration space in **§III.A**. The resulting orbital-space dimensions are a practical criterion for solver assignment. [QC-7, §3.2.1, Fig. 4; QC-3, §III.A]

5.2 Published demonstration: correlated solvers across a protein

Shajan *et al.* (2026) applied an embedded-wavefunction (EWF) workflow to the **folded and unfolded conformers of the Trp-cage protein**. After building localized fragment and interacting-bath orbitals, the authors organized embedded cluster Hamiltonians by their numbers of molecular orbitals. Their Figure 4 provides a visual explanation of why they divided the correlated calculations between FCI and sample-based quantum diagonalization (SQD). [QC-7, §§3.1.1, 3.2.1, original Fig. 4]

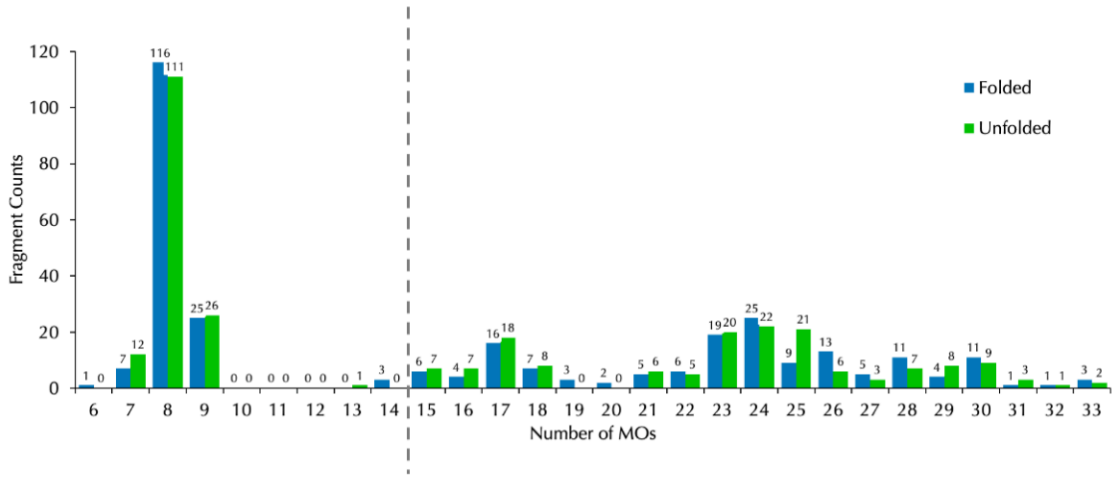


Figure 4: Number of EWF clusters with a given size (defined as the number of MOs) for the folded (blue) and unfolded (green) conformers. The black dashed line separates EWF clusters studied with FCI (left) and SQD (right).

Figure 8: Reproduced from Shajan et al. (2026), original **Fig. 4**, arXiv version 2, printed page **14** (PDF page **14**). The blue and green bars denote the numbers of EWF clusters of each orbital size in folded and unfolded **Trp-cage**, respectively. The dashed vertical boundary distinguishes the paper's FCI-treated clusters (through 14 molecular orbitals) from its SQD-treated clusters (15–33 molecular orbitals). This is **published Trp-cage data, not an output or histogram of the present MLN2238 calculation**. Full paper: arXiv:2512.17130v2.

The largest bars in Figure D.1 lie at **eight molecular orbitals**: the authors report **116** such clusters for folded Trp-cage and **111** for unfolded Trp-cage. They identify these pre-

dominantly with small hydrogen-atom environments and solve the small-cluster side of the distribution by classical FCI. At higher orbital counts, clusters become less numerous but more demanding; the reported range extends to **33 molecular orbitals**, and the 15–33 orbital group is assigned to SQD. [QC-7, §3.2.1, printed pp. 13–14, Fig. 4]

Figure 4 reports a distribution of **different cluster sizes**, and §3.2.1 explicitly assigns **FCI** to the smaller group and **SQD** to the larger group in this Trp-cage calculation. This source-reported allocation is the reason the figure appears here. [QC-7, §3.2.1, Fig. 4]

5.3 What is scientific about the size cutoff—and what it does not prove

The dashed line is the **FCI/SQD solver boundary used by the authors** in this specific calculation. The paper assigns clusters with fewer than 15 molecular orbitals to FCI and its 15–33-orbital clusters to SQD. The point is to adapt the solution method to the **computational size of the embedded Hamiltonian** while still producing correlated quantities needed for subsequent EWF reconstruction. [QC-7, §3.2.1]

Several distinctions matter when interpreting such a plot:

- The *number of molecular orbitals* measures an aspect of the configuration-space size; it is not itself the number of determinants, which also depends on the numbers of electrons in each spin sector.
- The papers do not equate **orbital count** with a ranking of correlation difficulty or biological influence.
- The selection of a cluster for **SQD** does not imply that quantum sampling alone returns its energy. SQD ultimately uses a classically solved Hamiltonian projected into a selected determinant subspace. [QC-6, Fig. 1, Results; QC-7, §2.2]

Thus, the figure is evidence for a **practical heterogeneous solver strategy** in the published Trp-cage calculation. It does not establish a unique cutoff transferable to every molecule or embedding construction.

5.4 Application: a different system and routing policy

The isolated-MLN2238 calculation uses the heterogeneous-solver approach for **42 fragment-plus-bath Hamiltonians**, rather than the published folded/unfolded Trp-cage systems. We are not comparing folded and unfolded Trp-cage conformers. [Challenge model definition; QC-7 is precedent, not a source for the MLN2238 results]

For the initial **classical baseline (Stage B)**, the established allocation is **19 FCI fragments and 23 CCSD fragments**. For the subsequent quantum-solver experiment, **Stage D identifies 11 candidate fragments by the implemented size rule** $n_{\mathrm{orb}} \geqslant 20$. The remaining fragments continue to receive classical treatment. After calibration, the intended full mixed campaign uses **19 FCI, 12 CCSD, and 11 TrimSQD** fragment solvers. These counts and the threshold describe **our Challenge configuration**, not numbers reported in the Shajan paper. [Challenge-defined validated stage allocation and routing]

The comparison can therefore be stated precisely:

	Shajan et al. (2026): published Trp-cage	**Proteasome Challenge: MLN2238**
Electronic problems	Embedded clusters from folded and unfolded protein conformers	42 embedded clusters of one isolated inhibitor molecule
Size-based boundary for quantum route	**More than 14 MOs** (SQD at 15–33 MOs)	**At least 20 MOs** for the 11 quantum candidates
Classical solver roles	FCI for the paper's small clusters; other classical references reported separately	Initial baseline: 19 FCI + 23 CCSD
Quantum-correlated method	SQD, with extended-SQD refinement in the published workflow	TrimSQD after a separate calibration experiment
Orbital-size distribution	Published **Figure 4** (folded/unfolded Trp-cage)	Distribution of the **42 calculated MLN2238 clusters**

The **20-orbital threshold** applies to the isolated MLN2238 cluster distribution and its solver allocation; it differs from the **15-orbital SQD threshold** in the published Trp-cage study. The two fragment populations and their histograms are distinct. [QC-7, §3.2.1; Challenge workflow definition]

5.5 Interpreting the MLN2238 fragment selection

The **computed orbital counts** determine which of the 42 MLN2238 clusters enter the quantum-sampled route. Under the implemented criterion, 11 fragments qualify. Their identities and orbital counts characterize the resulting quantum-candidate population, separately from the folded/unfolded Trp-cage data of Shajan et al. [Challenge workflow configuration; QC-7, Fig. 4]

This selection is a **size-driven candidate screen**, not a ranking of fragments by correlation difficulty or biological influence. The ability of quantum-selected determinants to describe a correlated state is examined separately during the single-fragment calibration. [QC-2; QC-6; inference and Challenge workflow]

5.6 From selection to calibration

Stage D does **not yet establish SQD accuracy**. It determines which embedded cluster Hamiltonians will be eligible for quantum-sampled treatment in a later calculation. In **Stage E**, the

Challenge studies a selected fragment (**fragment 9** in the established calibration configuration) and compares candidate SQD/TrimSQD settings before promoting a protocol to the full 42-fragment campaign. The selected protocol is then used in the later mixed-solver experiment. [Challenge-defined stage sequence; QC-6 SQD subspace principle]

This transition leads directly to the next scientific question: **given a cluster too expensive to solve by exhaustive FCI, can quantum measurements help identify a compact determinant subspace that still captures an accurate correlated energy?** That is the purpose of the sample-based quantum diagonalization and TrimSQD discussion that follows. [QC-6, Results; QC-8, TrimSQD methodology]

6 Calibration of Quantum-Sampled Correlated Solvers

6.1 From a selected fragment to a correlated wavefunction

The previous stage identified embedded molecular fragments whose orbital spaces make exhaustive determinant-based calculations increasingly expensive. Selecting a candidate, however, does not yet explain **how** its electronic correlation can be estimated. Sample-based quantum diagonalization (SQD) uses a quantum processor to propose many-electron *configurations* and a classical solver to recover an approximate correlated wavefunction in the configuration space those samples help identify. The final energy therefore comes from **projected Hamiltonian diagonalization**, not from treating the raw quantum measurements as an energy measurement. [QC-6, Results, Eqs. (2)–(3); QC-7, §3.2 and Fig. 3]

In the published Trp-cage protein study, Shajan *et al.* organized the fragment calculations into classical and quantum-treated routes. Their Figure 3 connects a size-dependent routing decision to quantum sampling, configuration recovery, classical diagonalization, and classical reference methods. In **that study**, clusters with more than 14 molecular orbitals were eligible for the SQD-based branch. This figure is a map of the **published Trp-cage workflow**, not a depiction of the current inhibitor demonstration. [QC-7, §3.2.1, original Fig. 3, preprint p. 13]

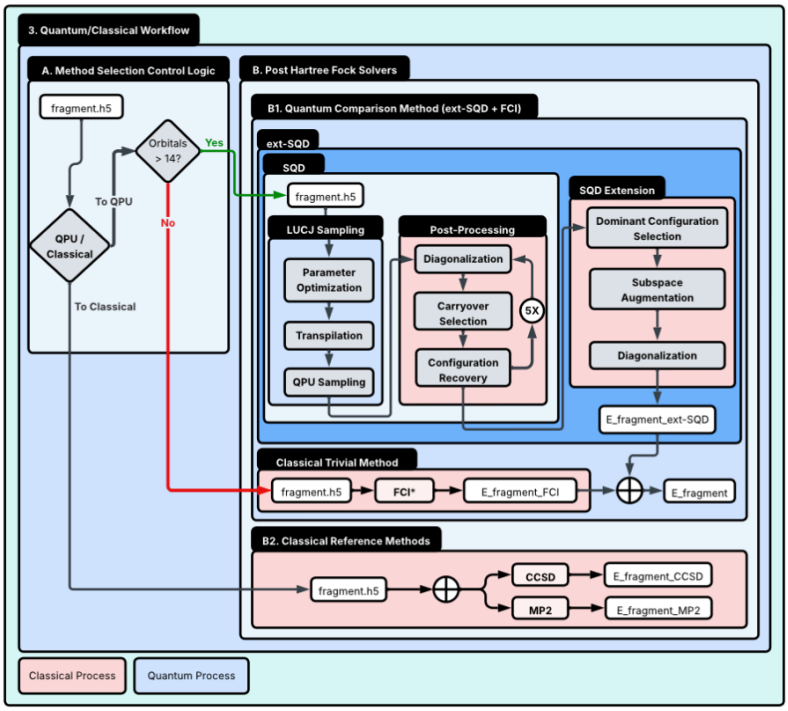


Figure 3: Quantum–classical workflow for post–Hartree–Fock EWF cluster calculations. The diagram summarizes solver selection logic, quantum circuit execution, and classical post-processing steps used in EWF-(SQD,FCI) calculations.

Figure 9: Original **Figure 3** from Shajan et al. (2026), "Molecular Quantum Computations on a Protein," arXiv:2512.17130v2, PDF/printed p. **13**. The routing boundary "Orbitals > 14?" belongs to the **published Trp-cage example**. Its SQD extension and classical reference paths establish the context for the SQD methods studied below. Journal DOI.

6.2 The SQD principle: sample configurations, then diagonalize

Consider an embedded molecular Hamiltonian represented in an orbital basis. In the occupation-number representation, a bitstring x identifies a Slater determinant $|x\rangle$; each spin-orbital is either occupied or empty. A prepared quantum circuit is measured repeatedly to yield a collection of such bitstrings, which are **candidates** for a useful determinant subspace. [QC-6, Results, Eq. (2)]

Robledo-Moreno et al. construct K **batches of** d **recovered configurations**, written $S^{(1)}, \ldots, S^{(K)}$. Their **Eq. (3)** defines the projected Hamiltonian separately for each batch:

$$\hat{H}_{S^{(k)}} = \hat{P}_{S^{(k)}} \hat{H} \hat{P}_{S^{(k)}}, \qquad \hat{P}_{S^{(k)}} = \sum_{x \in S^{(k)}} |x\rangle\langle x|.$$

The formulation uses the **batch-indexed subspaces** defined by Robledo-Moreno et al. The authors compute the ground states and energies by **iterative Davidson diagonalization**,

as stated immediately after Eq. (3). [QC-6, Results, Eq. (3), arXiv:2405.05068v3, PDF p. 4] Robledo-Moreno et al. explicitly contrast subspace selection and diagonalization with full configuration interaction, and solve for each batch state and energy using the Davidson procedure described after their Eq. (3). [QC-6, Results, Eq. (3) and following paragraph, PDF p. 4; QC-2 and QC-3 for FCI/CCSD background]

Real device measurements are noisy. A circuit expected to conserve the alpha and beta electron populations can yield measured strings in an incorrect particle-number sector. The **configuration recovery** procedure described by Robledo-Moreno *et al.* uses reference orbital occupations to probabilistically repair such strings. Recovered configurations are subsampled into batches; each batch defines a projected eigenproblem, and the resulting orbital occupations feed back into subsequent recovery iterations. The corrected strings therefore remain hypotheses about relevant configurations, not confirmed exact samples from the ideal wavefunction. [QC-6, Results pp. 3–5, Eqs. (2)–(4)]

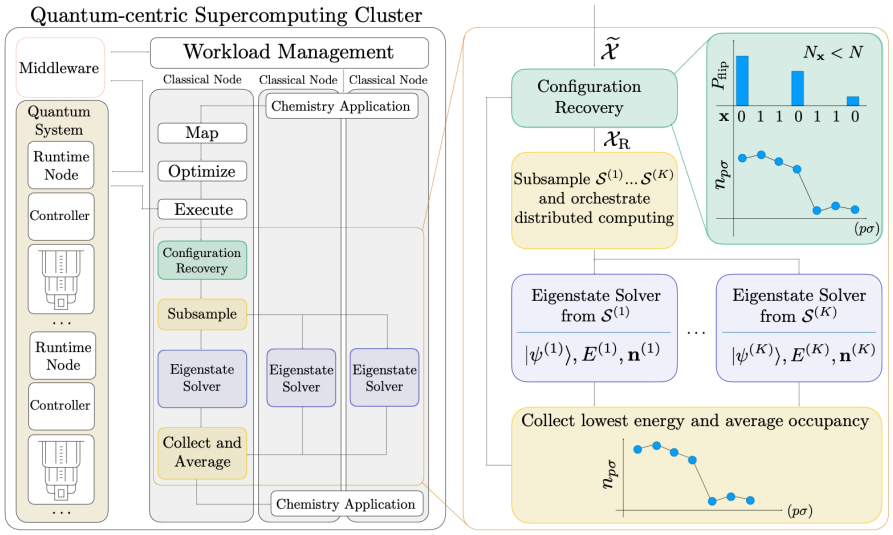


Figure 1: **Quantum-centric supercomputing architecture and sample-based quantum diagonalization workflow diagram. Left:** We illustrate a simplified architecture used to execute our workflow. The architecture has a cluster with a quantum system alongside classical runtime nodes within an isolated environment. A workload management system controls hybrid quantum-classical jobs through middleware. Our workflow is distributed on a set of classical nodes. It includes standard quantum chemistry application routines such as computing electronic integrals, mapping to qubits, and preparing circuits to be executed. **Right:** Details of the classical post-processing step. The input is a set of noisy samples $\widetilde{\mathcal{X}}$ from the quantum execution that are processed with our configuration recovery step, using information from a vector $\mathbf{n}$ of reference orbital occupancies. The green inset shows an example where a configuration with $N_{\mathbf{x}} < N$ is corrected. The set of recovered configurations $\mathcal{X}_{\text{R}}$ is sub-sampled and distributed for projection and diagonalization on parallel classical nodes. A new average reference occupancy vector $\mathbf{n}$ is computed from the results, and the configuration recovery loop is repeated self-consistently until convergence.

Figure 10: Original **Figure 1** from Robledo-Moreno et al. (2025), "Chemistry Beyond the Scale of Exact Diagonalization on a Quantum-Centric Supercomputer," arXiv:2405.05068v3, PDF p. **3**. The **right-hand panel** is the key scientific content here: noisy configurations $\widetilde{\mathscr{X}}$ undergo recovery, are divided into subspaces $S^{(k)}$, and enter classical eigensolvers; updated mean occupations guide the next round. The **left-hand panel** depicts the quantum-centric supercomputing architecture used in the published workflow. Published DOI.

In SQD, the **QPU supplies a data-dependent proposal for the important electronic configurations; the classical eigensolver supplies the correlated energy**. The authors report configuration recovery, subsampling, projected Hamiltonian diagonalization, and updated occupations as the consecutive steps of their method. [QC-6, Results, Fig. 1, Eqs. (2)–(4)]

6.3 Published SQD extensions and why selection quality matters

An initial recovered determinant set can miss energetically important configurations. **Extended SQD (ExtSQD)** improves the represented space by retaining influential configurations from earlier diagonalizations and augmenting the determinant subspace. In Shajan et al.'s published Trp-cage workflow, the extension follows the iterative SQD recovery and carryover loop, as shown in Figure E.1. [QC-7, §3.2, Fig. 3; QC-8, §IV.C and Fig. 2]

Merz *et al.* (2026) introduce **TrimSQD** as a refinement of this selection process for embedded correlated problems. Their Figure 2 sets ExtSQD and TrimSQD beside one another. Both begin with quantum sampling and use configuration recovery and carryover; TrimSQD introduces a **two-level diagonalization/selection** procedure. Configurations are first assessed within subgroups, then pooled and assessed through a further, combined diagonalization before extension of the selected subspace. This makes the comparison about **which configurations survive selection and jointly contribute to the correlated subspace**, not about replacing classical diagonalization with a quantum energy estimator. [QC-8, §IV.C.1, original Fig. 2, PDF pp. 5–6]

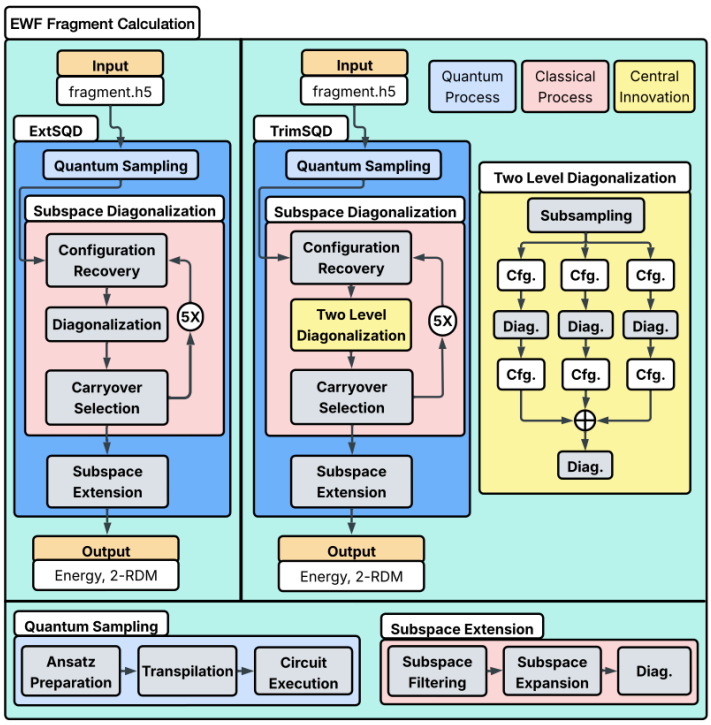


Fig. 2. The workflow of ExtSQD (left) and TrimSQD (middle), with central innovations from this study marked in light yellow (right). "Cfg." is an abbreviation for "electronic configurations", $\mathbf{x}$ in Eq. (1) of the main text.

Figure 11: Original **Figure 2** from Merz et al. (2026), "Crossing the 12,000-atom barrier with heterogeneous quantum-classical supercomputing: quantum chemistry of protein-ligand complexes," arXiv:2605.01138v2, PDF p. **5**. Left: published ExtSQD; center: published TrimSQD; right: detail of the two-level diagonalization/selection innovation. The light-yellow blocks identify the paper's innovations. The captioned "Cfg." denotes an electronic configuration.

Merz et al.'s reported TrimSQD results concern protein–ligand benchmark systems. The present MLN2238 calibration applies the TrimSQD method to a separate embedded molecular Hamiltonian. [QC-8, §IV.C.1; Challenge experiment]

6.4 Single-fragment calibration: MLN2238 fragment 9

The present calculation applies SQD/TrimSQD to **isolated MLN2238**, rather than to the folded/unfolded Trp-cage conformers or the protein–ligand complexes used in the published demonstrations. The embedding stage produced **42** embedded cluster Hamiltonians; the selection stage chose **11 quantum candidates with** $n_{\mathrm{orb}} \geqslant 20$. The 42-cluster count and the 20-orbital cutoff describe the present molecular calculation, not the published systems of Figures E.1–E.3. [Challenge Stage 2, Stage D, and Experiment A definitions]

A controlled **single-fragment calibration on fragment 9** compares quantum-sampling

and post-processing settings for *the same embedded Hamiltonian* before transferring the protocol to all 11 candidates. The test asks whether the candidate configurations and subspace-selection settings produce a stable, interpretable correlated energy at a tractable cost. Calibration must consider the electronic result and the numerical work needed to obtain it; a lower cost alone is not evidence of better correlation accuracy. [Challenge-defined Experiment A Stage E; QC-6, Eq. (3); QC-8, §IV.C.1]

The fragment-9 results characterize the energy obtained for the chosen fragment, changes with sampling or subspace-selection settings, the number and relevance of retained configurations, and comparison with the **available classical baseline for that same fragment**. FCI is exact **within its specified finite fragment space**; CCSD is a correlated approximation. Comparisons of the resulting energies concern the same embedded Hamiltonian and must distinguish FCI references from CCSD approximations. [QC-2, QC-3, QC-6]

The fragment-9 calibration selects the protocol identified as **`trim_800_r30`** for the subsequent quantum-routed clusters. The identifier denotes a solver configuration in this experiment; the correlated energy and numerical convergence are determined from the corresponding calculated results. [Challenge calibration configuration]

6.5 Interpret the result without overstating quantum advantage

The quality of the calibration depends on how consistently the selected determinant space represents the correlated result under the experiment's resource constraints. If two settings give similar energies, their respective determinant spaces and numerical costs can still justify choosing one over the other. Conversely, a favorable isolated-fragment energy comparison does not establish that every larger or chemically different fragment will converge under the same parameters. [QC-6, Results; QC-8, §IV.C; Challenge interpretation]

Most importantly, SQD and TrimSQD are **hybrid methods**. The role of the QPU is to supply information for selecting electronic configurations, whereas classical diagonalization recovers the wavefunction coefficients and correlated observables in the selected space. A single-fragment calibration does not on its own establish a quantum computational advantage or a clinical outcome. [QC-6, Fig. 1 and Eq. (3); QC-8, Fig. 2]

This calibration concerns the electronic Hamiltonian of an **isolated inhibitor fragment**. It is not an estimate of inhibitor–proteasome binding free energy, enzyme inhibition constants, or differences between two drug candidates. [Challenge scientific-model boundary]

6.6 From one protocol to the full correlated-fragment campaign

The scientific deliverable of Experiment A is a **selected correlated-solver protocol** that can be transferred to the quantum-routed embedded clusters. Once the method is fixed, the next experiment applies it across the selected fragments while preserving classical solvers on the other fragments. The planned mixed-solver allocation is **19 FCI, 12 CCSD, and 11 TrimSQD fragment problems**. Completing those calculations yields fragment-level correlated quantities that will later be assembled into a molecular electronic energy. [Challenge Experiment

B design; QC-7, §3.2; QC-8, §IV.A–C]

The calibrated protocol is subsequently applied to the selected quantum-routed fragments, alongside the classical solvers assigned to the remaining embedded problems.

7 The Full Embedded-Wavefunction Fragment Campaign

7.1 The published whole-molecule workflow

Shajan *et al.* (2026) study the **folded and unfolded conformers of the Trp-cage miniprotein** using an embedded-wavefunction (EWF) approach that combines quantum-sampled and classical correlated fragment solvers. Their Figure 1 presents the complete sequence. A Hartree–Fock reference is generated for each conformer; localized fragments and their initial entanglement baths are constructed and then expanded; the resulting EWF clusters are solved with correlated methods; finally, their outputs are used to reconstruct conformer energies. The figure labels **303 EWF cluster calculations** for the published application. This is the authors' Trp-cage calculation, not the inhibitor model used here. [QC-7, Fig. 1 and accompanying discussion, arXiv v2 pp. 5–7]

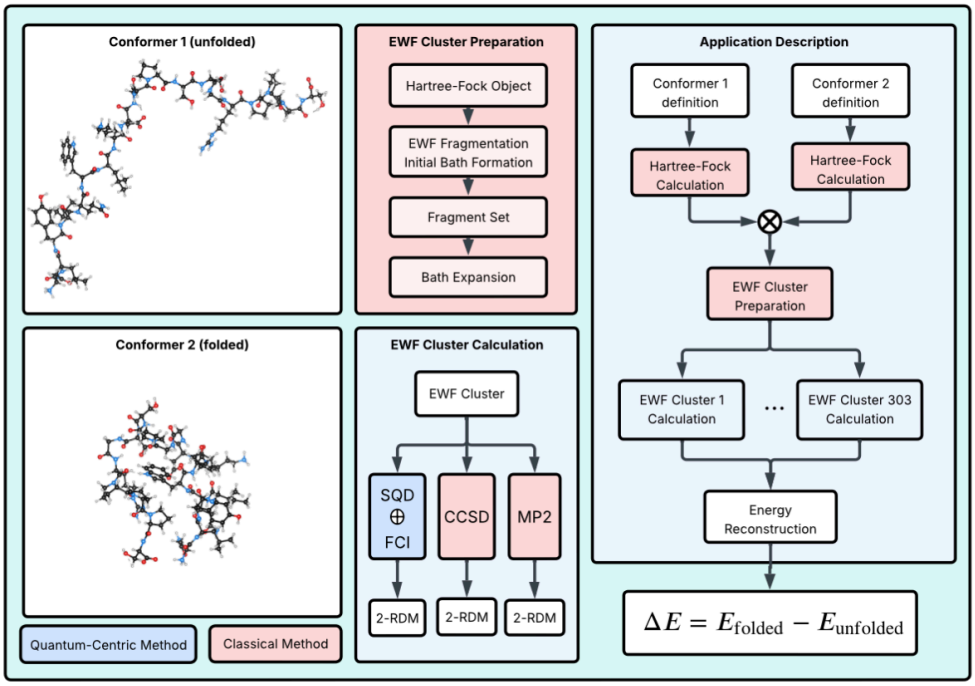


Figure 1: Overview of the EWF-based quantum-centric workflow for Trp-Cage conformer energetics. The protocol includes Hartree–Fock reference generation, automated fragmentation and bath construction, bath orbital expansion, and high-level cluster calculations using either quantum (SQD/FCI) or classical (MP2/CCSD) solvers. Fragment energies are reconstructed to obtain total conformer energies.

Figure 12: Original **Figure 1** from Shajan et al. (2026), "Molecular Quantum Computations on a Protein," arXiv:2512.17130v2, **PDF page 5**, DOI: 10.1021/acs.jctc.6c00364. The left-hand molecular drawings represent the **published Trp-cage conformers**; the workflow does **not** depict MLN2238. The paper's figure connects the full-molecule reference, fragment construction, correlated solvers, fragment density matrices, and whole-molecule energy reconstruction.

The paper's Methods section describes EWF fragments built from intrinsic atomic orbitals and their mean-field Schmidt baths, followed by an MP2-informed interacting bath expansion. Each final *EWF cluster* contains a fragment and its retained bath orbitals; the authors solve these embedded electronic problems separately. [QC-7, §§2.1.1–2.1.2, pp. 8–9; Fig. 2, p. 12]

7.2 What the published fragment calculations produce

In the Trp-cage work, the authors divide cluster calculations according to the **number of molecular orbitals**. Their Figure 4 and §3.2.1 identify **FCI for clusters with fewer than 15 molecular orbitals** and **SQD for clusters with 15–33 molecular orbitals**; CCSD and MP2 appear as classical comparison methods in the post-Hartree–Fock study. The original Figure 1 summarizes these solver pathways without specifying the finer details of the SQD recovery

loop, which are developed in §3.2.4 and Figure 8. [QC-7, §3.2.1, pp. 13–15, Figs. 3–4; §3.2.4, pp. 20–23]

The output required for reconstruction is not merely a list of fragment energies. Shajan *et al.* state that correlated solvers supply **one- and two-body reduced density matrices** for EWF clusters; their §3.3 describes projecting these quantities into the full molecular orbital space before assembling global expectation values. Their Figure 1 explicitly draws a **2-RDM** beneath each solver route. The mathematical reconstruction and handling of overlapping bath spaces are addressed in the next section, rather than being replaced here by a sum of isolated cluster energies. [QC-7, §2.1, p. 7; Fig. 1, p. 5; §3.3, p. 25]

The published application ultimately compares its folded and unfolded Trp-cage conformer energies. Figure 1 displays the **authors' own** conformer-difference expression:

$$\Delta E = E_{\text{folded}} - E_{\text{unfolded}}.$$

This equation is reproduced **from Shajan et al., Figure 1**. Their Table 1 reports relative conformer energies for EWF-MP2, EWF-CCSD, and EWF-(FCI,SQD). Those published values belong to the **two-conformer Trp-cage comparison**; they are not observable outputs of the single-molecule MLN2238 campaign. [QC-7, Fig. 1, p. 5; §4 and Table 1, pp. 25–26]

7.3 The related published protein–ligand demonstration

Merz *et al.* (2026) extend EWF and heterogeneous quantum–classical fragment solving to **protein–ligand systems**, including systems reported at **11,608 and 12,635 atoms**. Their method integrates quantum sampling and **TrimSQD** with classical computational procedures and EWF energy reconstruction. That publication evaluates fragment-level correlated solutions and, separately, whole-system and binding-energy results under its stated electronic-structure and basis-set approximations. It therefore provides the closest published reference here for mixed correlated fragment treatment involving TrimSQD, but its reported molecular systems and results are **not** those of this Challenge. [QC-8, abstract and §§II–IV, pp. 1–8; §VI.A and Table IX, pp. 8–9]

The published protein–ligand results concern different molecular systems. Correlated embedded-fragment energies and reconstructed whole-molecule or binding energetics are distinct quantities. [QC-7, §3.3, p. 25; QC-8, §VI.A, pp. 8–9]

7.4 Application: 42 embedded MLN2238 clusters

The present application of EWF treats **isolated MLN2238 (the active form of ixazomib) in STO-3G**. This is neither a Trp-cage conformer nor a complete proteasome–inhibitor complex. The fragment count and solver assignments below are **Challenge-specific configuration facts**, not results reported in QC-7 or QC-8. [Challenge Stage 1–2 / Experiment A–B definitions; published methodological precedent: QC-7, §§2.1, 3.2]

The established workflow prepares **42 embedded cluster problems**. Experiment A first

establishes a classical baseline with **19 FCI** and **23 CCSD** fragment calculations; Stage D identifies **11 quantum candidates** with the implemented size rule **at least 20 molecular orbitals**. Stage E calibrates the quantum-sampled method on **fragment 9** and promotes the designated `trim_800_r30` solver configuration. Experiment B then applies the mixed assignment:

Correlated solver	Assigned MLN2238 clusters	Role in the calculation
FCI	**19**	Classical correlated treatment of the smaller assigned clusters
CCSD	**12**	Classical correlated treatment retained for other clusters
TrimSQD	**11**	Quantum-sampled correlated treatment with the calibrated protocol
Total	**42**	One correlated result expected for every embedded cluster

Table 06.1. Solver allocation for the 42 embedded MLN2238 clusters in the present experiment.

These allocations differ from the published Trp-cage FCI/SQD assignment and the protein–ligand studies of Merz et al.; the calculated molecular systems are distinct. [Challenge-defined solver allocation; QC-7, §3.2.1, Fig. 4; QC-8, §§IV–V]

7.5 Reading the full-campaign outputs

The EWF campaign yields correlated fragment solutions characterized by fragment identity, orbital-space size, assigned solver, computed energy, and density-matrix data required for molecular reconstruction. The calculated MLN2238 quantities are distinct from the published Trp-cage fragment results. [QC-7, §3.2.1 and §3.3, pp. 13, 25; Challenge-defined output inspection]

For this demonstration, the **42/42 completed-result checkpoint** is a requirement for proceeding to the final reconstruction stage. It means verifying that every expected embedded cluster has an acceptable, correctly associated correlated result and required downstream data; it does **not** by itself prove that TrimSQD has reproduced FCI quality or that the molecular electronic energy has converged with embedding parameters. The numerical evidence for either stronger conclusion would require the appropriate comparisons and recorded results. [Challenge validation criterion, distinct from published claims]

7.6 Transition to molecular reconstruction

After all fragment calculations, the published Trp-cage EWF study **projects the cluster one- and two-body density matrices into the full molecular orbital basis** and combines their contributions to compute whole-molecule expectation values. The authors specifically discuss symmetrization over overlapping bath contributions. [QC-7, §3.3, p. 25] The detailed reconstruction formalism and its limitations are the subject of Nusspickel, Ibrahim, and Booth (2023). [QC-5, §§II.A–II.B, Eqs. 12–18]

Reconstruction of molecular energy from the correlated cluster density matrices follows the embedding methods described by Nusspickel, Ibrahim, and Booth (2023) and Shajan et al. (2026). The resulting isolated-inhibitor energy is distinct from a measured proteasome binding affinity.

8 Molecular Energy Reconstruction

8.1 Why separate fragment calculations require reconstruction

The preceding experiment solves correlated electronic problems for many **fragment-and-bath clusters**. These local wavefunctions do not, by themselves, constitute a whole-molecule electronic energy. Nusspickel, Ibrahim, and Booth (2023) study the reconstruction of **global expectation values** from *ab initio* quantum-embedding solutions and show that the choice of reconstruction functional changes the answer even when the individual embedded wavefunctions are held fixed. Their introduction and Figure 1 make this distinction central to the method. [QC-5, Introduction and Figure 1, PDF pp. 1–3]

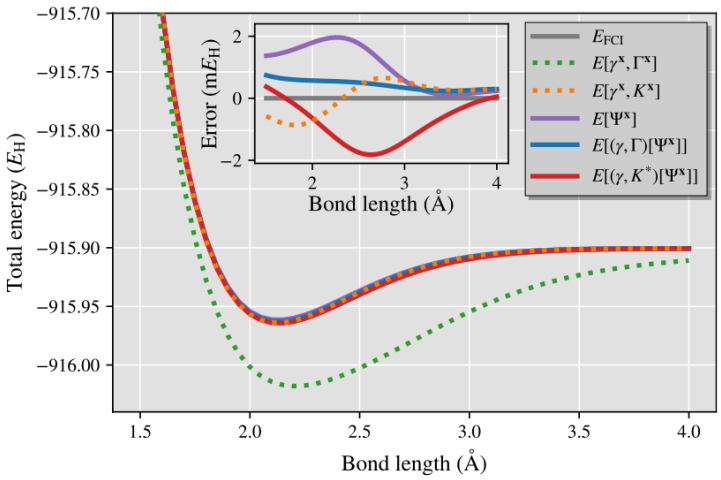


Figure 1. Binding energy of the chlorine dimer in a STO-6G basis compared to FCI, with $E[\gamma^x, \Gamma^x]$ corresponding to the democratically partitioned energy expression of traditional DMET.[33] Different energies correspond to different total energy functionals, constructed from the same solutions to the two DMET embedding problems (with fragment spaces of the (Löwdin-orthogonalized) core and valence orbitals of each chlorine atom). The two cluster spaces are defined by 10 orbitals (9 fragment and 1 bath), with the full space comprising 18 orbitals. The motivation and definition of these different energy functionals from the embedded wave functions are given in the rest of this work, with more details on this system (and others like it) discussed in Sec. 4.

Figure 13: Different energies from the same embedded solutions. The authors calculate the chlorine-dimer energy curve in STO-6G using two atomic DMET clusters and compare several reconstructed energy functionals against FCI. The inset reports differences in millihartree. The cluster spaces and their solved wavefunctions are the **same** across the curves; what differs is how their contributions are combined. This is a diatomic chlorine example, **not** an inhibitor-binding calculation. [QC-5, Figure 1 and accompanying discussion, PDF pp. 2–3]

8.2 Recombining reduced density matrices

In §II.A, the authors first explain the standard **democratic partitioning of reduced density matrices**. Each cluster solution produces a one-body density matrix and a two-body density matrix. They write the reconstructed density matrices as sums of fragment-projected cluster contributions in their original **Eqs. (1) and (2)**: [QC-5, §II.A, PDF p. 3]

$$\gamma_{pq} = \sum_{\mathbf{x}}^{N_{\mathrm{frag}}} (\hat{P}^{\mathbf{x}}\gamma^{\mathbf{x}})_{pq} \tag{1}$$

$$\Gamma_{pqrs} = \sum_{\mathbf{x}}^{N_{\mathrm{frag}}} (\hat{P}^{\mathbf{x}}\Gamma^{\mathbf{x}})_{pqrs}, \tag{2}$$

Here the symbol x labels the fragment and its associated cluster, and the operator P-hat with superscript x projects the contribution onto that fragment's orbital subspace. The definitions of the underlying cluster density matrices appear as Eqs. (3) and (4) in the paper. Their symmetrized projection is specified in **Eqs. (5) and (6)**. The projection is necessary because the bath spaces attached to different fragments can overlap even when the assigned fragment spaces do not. [QC-5, §II.A, PDF pp. 3–4, Eqs. (3)–(7)]

The paper then writes an energy functional for these democratically partitioned cluster density matrices, including the effective core-Hamiltonian contribution from doubly occupied orbitals outside the active cluster. This is the source's **Eq. (9)**: [QC-5, §II.A, PDF p. 4]

$$E[\gamma^{\mathbf{x}}, \Gamma^{\mathbf{x}}] = E_{\mathrm{nuc}} + \sum_{\mathbf{x}}^{N_{\mathrm{frag}}} \left[\sum_{pq}^{N_{\mathrm{cl}}^{\mathbf{x}}} \tilde{h}^{\mathbf{x}}_{pq} (\hat{P}^{\mathbf{x}}\gamma^{\mathbf{x}})_{qp} + \frac{1}{2} \sum_{pqrs}^{N_{\mathrm{cl}}^{\mathbf{x}}} (pq|rs)(\hat{P}^{\mathbf{x}}\Gamma^{\mathbf{x}})_{pqrs} \right], \tag{9}$$

The authors specify conditions under which this construction becomes exact: the fragment projectors must cover the relevant full orbital space and the cluster density matrices must become exact in the complete-cluster limit. Away from those limits, they explain that democratically partitioned density matrices do **not generally satisfy N-representability**—the condition of corresponding to an admissible global fermionic wavefunction. [QC-5, §II.A, PDF pp. 4–5, Eq. (10) and surrounding discussion]

8.3 The published cumulant alternative

In §II.B, the paper develops a different route: partitioning the **two-body cumulant** rather than the full two-body density matrix. The authors define the cumulant decomposition in the spin-restricted convention using their **Eq. (12)**: [QC-5, §II.B, PDF p. 5]

$$\Gamma_{pqrs} = \gamma_{pq}\gamma_{rs} - \frac{1}{2}\gamma_{pr}\gamma_{sq} + \tilde{K}_{pqrs}. \tag{12}$$

In the authors' explanation, the products of the one-body density matrix contain the factorizable or disconnected contributions to the two-body density matrix. The cumulant retains the additional two-body part, while reconstructing those disconnected products from the combined one-body density matrix allows contributions involving **different fragments** to enter the estimate. This is the stated rationale for the cumulant partitioning in §II.B. [QC-5, §II.B, PDF p. 5, Eqs. (12)–(14)]

In the calculations reported by Nusspickel et al., the exact cumulant definition in Eq. (12) is not used unchanged. It decomposes the correlated one-body density matrix into the mean-field reference plus its correlated change and adopts a **modified approximate cumulant**, written in its **Eq. (15)**. The authors explicitly relate this approximation to the true cumulant in Eq. (16). [QC-5, §II.B, PDF p. 6]

> 1-RDM can be decomposed as $\gamma = \gamma^{\upsilon} + \Delta\gamma$, where γ^{υ} is the reference mean-field density-matrix. We then work with a slightly modified cumulant definition, K, as
>
> $$\begin{aligned}\Gamma_{pqrs} &= \gamma^0_{pq}\gamma^0_{rs} + \gamma^0_{pq}\Delta\gamma_{rs} + \Delta\gamma_{pq}\gamma^0_{rs} \\ &\quad - \frac{1}{2}(\gamma^0_{pr}\gamma^0_{sq} + \gamma^0_{pr}\Delta\gamma_{sq} + \Delta\gamma_{pr}\gamma^0_{sq}) + K_{pqrs}.\end{aligned} \tag{15}$$

The resulting energy reconstruction used for the paper's partitioned-cumulant results is **Eq. (18)**. It combines the reference Hartree–Fock energy, a correction formed from the reconstructed one-body density change and the reference Fock matrix, and contributions of the fragment-projected approximate cumulants. [QC-5, §II.B, PDF p. 6]

$$\begin{aligned}E[\gamma^{\mathbf{x}}, K^{\mathbf{x}}] &= E_{\mathrm{HF}}[\gamma^0] + \sum_{pq} F_{pq}[\gamma^0]\Delta\gamma_{qp} \\ &\quad + \frac{1}{2}\sum_{\mathbf{x}}^{N_{\mathrm{frag}}}\sum_{pqrs}^{N^{\mathbf{x}}_{\mathrm{cl}}}(pq|rs)(\hat{P}^{\mathbf{x}}K^{\mathbf{x}})_{pqrs},\end{aligned} \tag{18}$$

Equations (9) and (18) define **different energy-reconstruction functionals** based on distinct partitions of the embedded density-matrix information. [QC-5, §II.A–B, PDF pp. 4–6]

8.4 What the nitrogen dissociation comparison demonstrates

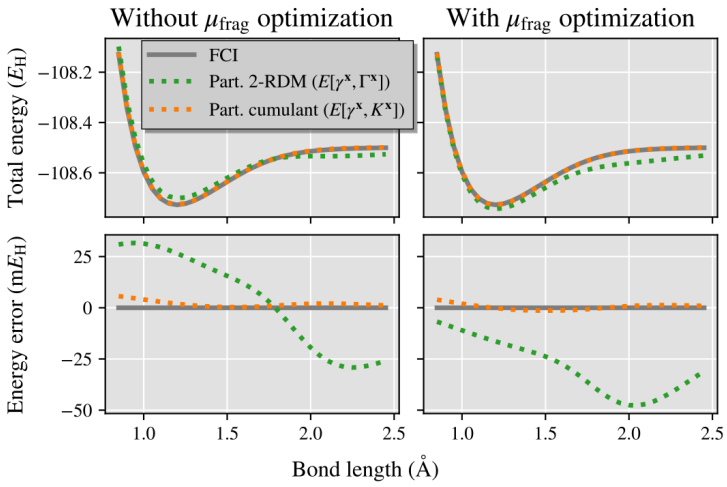


Figure 2. Dissociation curve of N_2 in the STO-6G basis, calculated from two atomic DMET embedding problems. The same DMET-FCI cluster solution is used in both the partitioned density matrix (Sec. 2.1) and partitioned cumulant (Sec. 2.2) energy expressions, $E[\gamma^x, \Gamma^x]$ and $E[\gamma^x, K^x]$ respectively, defined as given by eqs 9 and 18. Plots on the right have optimized a global chemical potential to ensure that the democratically partitioned 1RDM traces to the correct number of electrons, while the plots on the left omit this chemical potential optimization.

Figure 14: N_2 dissociation in STO-6G. Both reconstruction approaches use the **same DMET-FCI cluster solutions** for two atomic embedding problems. The columns compare results **without and with** fragment chemical-potential optimization; the top panels show the energy curves and the lower panels their errors against FCI. In this test, the cumulant-based result remains markedly closer to the FCI energy over the displayed bond-length range than direct partitioning of the two-body density matrix. The accompanying text emphasizes that the cumulant-partitioned result changes relatively little under chemical-potential optimization in this example. [QC-5, Figure 2 and §II.B, PDF pp. 5–6]

The authors also examine reconstruction directly from cluster wavefunctions in **§III**, rather than from democratically partitioned density matrices alone, as an alternative motivated in part by N-representability. Consequently, Figure 2 illustrates a comparison **among published reconstruction functionals**; it does not establish that all schemes have the same representability properties or that one is universally exact at finite cluster size. [QC-5, §§II.A–III, PDF pp. 4–7]

8.5 How reconstruction appears in the published EWF protein study

Shajan *et al.* (2026) describe the final step of their published Trp-cage calculation in **§3.3, "Reconstruction of Global Energies from EWF Clusters."** Their fragment solvers supply one- and two-body reduced density matrices in the embedded spaces. The authors report projecting these back into the full molecular space and symmetrizing the contributions associated with overlapping bath orbitals before evaluating their full-system energies. Their work then compares **folded and unfolded conformers**, for which their Figure 1 and Table 1 report the corresponding energetic comparison. [QC-7, §3.3, arXiv v2 PDF p. 25; Figure 1 p. 5; Table

1 p. 26]

This provides the direct published bridge from fragment-level FCI/SQD solutions to **a molecular energy or energy difference**. The reconstruction study by Nusspickel *et al.* explains why the assembly of fragment information requires care, and why alternatives based on cumulants or more global wavefunction constructions have been investigated. [QC-5, §§II–III; QC-7, §3.3]

8.6 Application: reconstructing isolated MLN2238

The present calculation treats **one isolated MLN2238 molecule in STO-3G**, partitioned into **42 EWF embedded clusters**. The mixed correlated calculation assigns **19 clusters to FCI, 12 to CCSD, and 11 to TrimSQD** before molecular reconstruction. [Challenge computational configuration; QC-7, §3.3]

The molecular-reconstruction calculation combines available correlated-cluster information into full-molecule electronic quantities. Its principal observables are the reconstructed molecular energy, any reported correlation correction, and comparison with the classical baseline calculated for the same molecular system. [QC-5, §II; QC-7, §3.3; Challenge calculation]

Nusspickel et al.'s cumulant energy expression, Eq. (18), describes a specific published reconstruction functional. The numerical molecular energy also depends on the reconstruction functional, density-matrix conventions, and finite embedded spaces used in the calculation. [QC-5, Eqs. (9), (12), (15), (18)]

8.7 Scientific interpretation and limitations

The two source studies give different levels of comparison. Nusspickel *et al.* benchmark **reconstruction prescriptions** against FCI in model molecular systems, while Shajan *et al.* reconstruct **energies of two Trp-cage conformers** using embedded correlated solutions. The present calculation reconstructs the **electronic energy of a single isolated inhibitor** within its finite orbital basis and embedded fragment space. Those are related applications of embedding, but **they are not the same measured quantity**. [QC-5, Figures 1–2; QC-7, §3.3 and Table 1; Challenge model definition]

Interpretation must therefore be limited by the recorded evidence and source-described approximations: a finite orbital basis, finite cluster and bath spaces, the chosen correlated solvers, and the selected energy-reconstruction prescription. A completed 42/42 fragment campaign establishes coverage of the assigned embedded calculations; it does **not** on its own establish convergence of the global energy, exactness of the selected wavefunctions, or agreement with biological inhibition measurements. [QC-5, §II.A–B, pp. 4–6; QC-7, §§3.2–3.3; Challenge validation criterion]

Most importantly, this experiment **does not calculate a bortezomib-versus-ixazomib binding free-energy difference**, and its isolated MLN2238 electronic energy is **not** an estimate of an experimental β5 inhibition constant. Such comparisons belong to a separate,

explicitly constructed proteasome–ligand energetic study. [Challenge model scope; QC-7, §4; QC-8, protein–ligand modeling scope]

The overall scientific sequence is now complete: *mean-field reference → fragment and bath construction → correlated fragment solvers → quantum-sampled calibration and full campaign → reconstruction and bounded interpretation of the molecular energy.*

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